

CSS — Lecture 09 Summary Notes

Coding and Web Development | By Nishant Sir

1. What is CSS?

CSS = Cascading Style Sheets

CSS is used for **formatting and styling** the appearance of a webpage.

HTML vs CSS (Real-life Analogy)

	HTML	CSS
Role	Structure (Skeleton) of a page	Style (Looks & Design) of a page
House Analogy	Foundation, pillars, walls, roof, doors, windows	Paint, furniture, lighting, tiles, decoration
Result	Structure is complete but looks plain	Same structure, but looks completely different

Example:

```
<body>
  <h1>Welcome to My House</h1>
  <p>This is my simple house.</p>
</body>

body { background-color: #f5f5f5; font-family: Arial; }
h1 { color: #2c3e50; text-align: center; }
p { color: #555; font-size: 18px; line-height: 1.6; }
```

Summary: HTML builds the **structure**. CSS **styles** and beautifies it. Together, they build a complete webpage.

2. Types of CSS

There are **3 types** of CSS:

1. Inline CSS
2. Internal CSS
3. External CSS

a) Inline CSS

Applied directly inside an HTML tag using the style attribute — used for styling a **single element**.

```
<p style="color: red;">Hello</p>
```

b) Internal CSS (Embedded CSS)

Written inside the same HTML file, but placed inside a <style> tag within the <head> section.

```
<head>
```

```
<style>
  p { color: blue; }
</style>
</head>
```

c) External CSS

Styles are written in a **completely separate .css file**, and linked to the HTML file using the <link> tag.

```
<link rel="stylesheet" href="index.css">
```

Diagram:

HTML file <---- <link rel="stylesheet" href="index.css"> ----> CSS
file (index.css)

3. CSS Selectors

Selectors decide **which HTML element** will be styled.

Syntax:

```
selector {
  property: value;
}
```

a) Element (Type) Selector

Selects HTML elements directly by their **tag name**.

```
p {
  color: red;
}
```

b) Class Selector

Targets elements having a specific **class** attribute. Defined using a **period (.)** followed by the class name. Same class can be applied to multiple elements.

```
<p class="hello">Hello</p>

.hello {
  color: blue;
}
```

c) ID Selector

Targets a **single, unique** element based on its id attribute. Defined using a **hash (#)** followed by the id name.

```
<p id="sagar">Hello</p>

#sagar {
  color: green;
}
```

d) Universal Selector

Selects **every** element on the page — acts as a wildcard (*).

```
* {
  margin: 0;
  padding: 0;
}
```

}

4. CSS Specificity

Specificity decides which rule “wins” when multiple selectors target the same element.

Order (highest → lowest priority):

Inline CSS > ID > Class > Element > Universal (*)

5. CSS Combinators

Combinators define the **relationship** between elements (based on Parent → Child structure).

Combinator	Symbol	Meaning
Descendant	A B (space)	B is inside A, at any depth
Child	A > B	B is a direct child of A
Adjacent Sibling	A + B	B is immediately next to A (same parent)
General Sibling	A ~ B	B comes after A anywhere among siblings (same parent)

a) Descendant Combinator (space)

Selects all elements nested inside a specified ancestor, no matter how deep.

```
div p {  
  color: red;  
}
```

b) Child Combinator (>)

Selects only **direct/immediate** children of an element.

```
div > p {  
  color: red;  
}
```

c) Adjacent Sibling Combinator (+)

Selects the element that comes **immediately after** another, sharing the same parent.

```
h1 + p {  
  color: red;  
}
```

d) General Sibling Combinator (~)

Selects **all** matching siblings that come after a specified element (same parent).

```
h1 ~ p {  
  color: red;  
}
```

6. Pseudo-classes vs Pseudo-elements

Pseudo-class

A keyword added to a selector that defines a **special state** of an element.

Syntax:

```
selector:pseudo-class {  
  property: value;  
}
```

Example: :hover, :nth-child()

Pseudo-element

A keyword added to a selector that lets you style a **specific part/content** of an element.

Syntax:

```
selector:::pseudo-element {  
  property: value;  
}
```

Example: ::before, ::after

Quick Comparison

	Pseudo-class	Pseudo-element
Syntax	Single colon :	Double colon ::
Targets	Element's state	Element's part/content
Example	:hover, :nth-child	::before, ::after

🔑 Key Takeaways

- CSS has **3 types**: Inline, Internal, External
- **4 basic selectors**: Element, Class, ID, Universal
- **Specificity order**: Inline > ID > Class > Element
- **4 combinators**: Descendant (space), Child (>), Adjacent Sibling (+), General Sibling (~)
- **Pseudo-class** (:) = state-based | **Pseudo-element** (::) = part/content-based